Executive Summary – Group 20
Does fine dust (PM2.5) reduce Seoul's Ttareungyi bike rentals?

<Hypotheses and data>
H_0: Higher daily PM2.5 is not associated with lower rentals. / H_1: Higher PM2.5 is associated with lower rentals (one-sided). Pre-specified follow-up: leisure-riding proxies (Han River station share, round trips) respond more on days off. Data: 727 days in 2024–2025 (80,803,663 rentals, 2,904 stations; Seoul Open Data Plaza, KMA ASOS 108). "Bad" means PM2.5 > 35 μg/m³ (58 days). Regression standard errors are Newey-West (HAC, 7 lags); DML uses 5-fold cross-fitting with cross-fitted standard errors.

<Hypothesis tests: the raw gap disappears after controls>

Hypothesis (H_0 vs. H_1)	Test	Result
Bad days: same rentals vs. fewer	Log-OLS	−10.5% (95% CI −21.1 to +1.6); one-sided p = 0.043 (two-sided 0.086) → significant before controls
Same, with weather and calendar controls	Multiple regression	+3.2% (95% CI −1.3 to +7.9); p = 0.168 → H_0 not rejected
PM2.5 per +10 μg/m³: no effect vs. effect	Double Machine Learning	RF −0.32% (−1.83 to +1.21), GB +0.29% → H_0 not rejected
Leisure proxies (Han River share, round trips) respond more on days off	DML, days off vs. workdays	Difference p = 0.53–0.79 (Han River share), 0.75–0.95 (round trips) → H_0 not rejected

<Why the raw gap appears: bad air falls on cold days>
51 of the 58 Bad days fell in winter or spring (7.8°C vs. 16.1°C on Good days), and winter rentals ran about 50% below the other seasons. Comparing within the same month flips the gap from −10.5% to +19.4%. Weather, season, calendar and trend explain 85% of daily rentals, with temperature and rain carrying most of the models' predictive weight (58% and 22%). After these factors are accounted for, the estimated PM2.5 effect is close to zero, with the 95% CI within about ±2% per +10 μg/m³.

<Exploratory finding (not pre-specified)>
On workdays, the Han River station share of rentals fell by 1.17% per +10 μg/m³ (95% CI −1.75 to −0.58; about 5.37% → 5.31%). It held in every robustness check and survives a Bonferroni correction across 16 core tests ($p < 0.001$). It is a relative-share change only: absolute Han River rentals showed no detectable change (−0.8%, p = 0.43).

<Implications>
Plan bike supply primarily around weather and season rather than daily PM2.5 readings alone, and do not cite the raw −10.5% gap as evidence that fine dust deters cycling. In a what-if (an assumption, not a measurement), losing about 5,200 bike-km/day (our upper-bound estimate per +10 μg/m³) with 30% replaced by car trips would add about 330 $kgCO_2eq$/day (0.2111 $kgCO_2eq$/passenger-km; Ministry of Environment & KECO, 2024).

<Limitations>
Observational city-wide daily totals (commute hours not visible); only 18 Bad days off; PM10 alerts on only 7 days; weather from one station.

Webpage: siwoo9573.github.io/ACDT-Group20-PM25-Ttareungyi · Code and data: github.com/siwoo9573/ACDT-Group20-PM25-Ttareungyi